DATA COLLECTION

70K+ MRI, 30Kh EEG

UK Biobank
50,000, 45-82y

ABCD Study
12,000, 9-14y

HCP Dev+Aging
2,500, 5-100y

Temple EEG
30,000h, All ages

CHA Hospital
3,000, 8-25y

Chosun Univ
10,000, 50-90y

PREPROCESS

MRI	FreeSurfer fMRIPrep
EEG	Artifact 200Hz
Genomic	QC+Impute PRS

PRETRAINING: 50B

fMRI FM	15B	4D Swin MAE
EEG FM	15B	DIVER-XL
Genomic FM	15B	PRS-T
Q-Former	500M	Cross-Modal

TRAJECTORY

(Killer Task)

CNF Model
Individual Trajectory

Dev Traj
CHA 3K Longitudinal

Aging Traj
Chosun 10K Longitudinal

VALIDATION

Internal (UK Biobank 20%)
Hold-out test set, cross-validation

Clinical Pilot (2K)
Real-world deployment

Clinician feedback loop

Korean Validation (13K)
CHA 3K (Dev) + Chosun 10K (Aging)

Domain adaptation, ethnicity bias check

DEPLOYMENT

Knowledge Distillation
50B -> 3B

INT8 Quantization
Optimized inference

API Service
<30sec, 8 tasks

8 DOWNSTREAM TASKS

- Dev Trajectory r>0.5
- Aging Trajectory r>0.5
- AD Diagnosis AUC>0.90
- PD Diagnosis AUC>0.85
- Zero-shot Acc>70%
- Few-shot AUC>0.75
- Dementia Conv HR>2.0
- Brain Age MAE<2.5y

MILESTONES & GO/NO-GO GATES

Milestones:

M1 (Y2)	M2 (Y3)	M3 (Y4)	M4 (Y5)
MAE<2.5y, Retrieval>80%	Dev Traj r>0.5, AUC>0.80	Dementia AUC>0.85	Latency<30s, Avg AUC>0.85

Go/No-Go Gates:

G1	G2	G3	G4	G5
N>50K, 3B done	MAE<3.0y	Dev Traj r>0.4	Aging AUC>0.80	Inference<60s

Figure 3. LifeSpan-FM: Data Pipeline and Training Workflow